# A Non-Singular $j$-Algebraic Scalar-Tensor Framework: Resolving Galactic Rotation, Black Hole Horizon Singularities, and the Hubble Tension

Advait Patel[1,*]

[1] *Independent Researcher*

(Dated: July 12, 2026)

We present an extended scalar-tensor theory of gravity augmented by a 4D nilpotent division algebra ($j$-algebra where $j^5 = 0$). By introducing a non-singular spatial horizon boundary condition ($g_{rr} = j$), coordinate singularities at black hole cores and event horizons are regularized. From a unified Action principle featuring a non-linear kinetic term $K(X)$ and a density-dependent potential $V(\psi)$, we derive field equations that account for flat galactic rotation curves without dark matter particles, predict a 3D photon sphere shadow diameter ($\theta \approx 41.3\,\mu$as) matching Event Horizon Telescope observations of M87*, and resolve the $5\sigma$ Hubble Tension by driving a local vacuum expansion boost ($H_{\text{local}} \approx 73.0$ km/s/Mpc). Finally, we demonstrate that a density-dependent screening mechanism suppresses the scalar field in high-density environments, recovering standard General Relativity within the Solar System and binary pulsar regimes down to $\mathcal{O}(10^{-9})$.

## I. INTRODUCTION & $j$-ALGEBRA AXIOMS

Standard General Relativity (GR) and the ΛCDM cosmological model face fundamental challenges at extreme limits:

1. **Curvature Infinities:** The Schwarzschild metric predicts non-physical singularities where curvature invariants diverge to infinity ($\infty$) at $r = 0$.
2. **The Dark Sector:** Explaining galactic dynamics and cosmic acceleration requires introducing unseen Dark Matter and Dark Energy.
3. **The Hubble Tension:** A persistent $5\sigma$ discrepancy exists between early-universe CMB measurements ($H_0 = 67.4$ km/s/Mpc) and local distance ladder measurements ($H_0 = 73.0$ km/s/Mpc).

To address these boundary issues simultaneously, we define an algebraic extension to real space $\mathbb{R} \to \mathbb{R}[j]$ by introducing the generator $j$, representing a regulated boundary operator:

$$j \equiv \lim_{\epsilon \to 0^+} \frac{\epsilon}{\epsilon^2 + \delta_j} \tag{1}$$

To prevent infinite algebraic expansion at higher tensor orders, $j$ obeys a **4D Nilpotency Condition**:

$$j^n \neq 0 \quad \text{for } n < 5, \qquad j^5 = 0 \tag{2}$$

Under this algebraic boundary condition, any smooth function $f(x)$ evaluated near a singular point $x_0$ terminates as a finite 4th-order expansion:

$$f(x_0 + j) = f(x_0) + jf'(x_0) + \frac{j^2}{2!}f''(x_0) + \frac{j^3}{3!}f'''(x_0) + \frac{j^4}{4!}f^{(4)}(x_0) \tag{3}$$

In Schwarzschild coordinates, the radial metric component near singular boundaries $r \leq r_s$ is mapped as:

$$g_{rr}^{(j)} = \left(1 - \frac{r_s}{r} + j\right)^{-1} \tag{4}$$

At the horizon ($r = r_s$), the metric component evaluates to $g_{rr}^{(j)}(r_s) = j^{-1}$, giving a finite inverse metric component $g^{rr}_{(j)} = j$. This regularizes the horizon geometry into a well-defined manifold.

## II. THE UNIFIED ACTION & DERIVED FIELD EQUATIONS

The complete dynamics of spacetime and the scalar field originate from a single invariant Action principle ($S$):

$$S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{tax}}(\psi, \nabla_\mu\psi)\right] \tag{5}$$

The scalar field Lagrangian density $\mathcal{L}_{\text{tax}}$ is defined as:

$$\mathcal{L}_{\text{tax}} = X + K(X) - V(\psi) \tag{6}$$

where $X \equiv -\frac{1}{2}g^{\mu\nu}\nabla_\mu\psi\nabla_\nu\psi$ is the standard kinetic term, $K(X)$ represents a non-linear kinetic self-interaction, and $V(\psi)$ is the density-dependent interaction potential:

$$V(\psi) = \frac{c^4\psi^2}{2kM_{\text{collective}}}\left(e^{\frac{\pi c T_{\text{local}} d(\hat{\Omega})}{kM_{\text{collective}}}} - 1\right) \tag{7}$$

### A. Master Equation I (Spacetime Geometry)

Varying the total Action with respect to the inverse metric tensor ($\frac{\delta S}{\delta g^{\mu\nu}} = 0$) yields the modified Einstein Field Equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{tax}}\right) \tag{8}$$

* your.email@example.com

where the field stress-energy tensor $T^{\text{tax}}_{\mu\nu}$ is given explicitly by:

$$T^{\text{tax}}_{\mu\nu} = (1 + K'(X))\,\nabla_\mu\psi\nabla_\nu\psi - g_{\mu\nu}\,(X + K(X) - V(\psi)) \tag{9}$$

### B. Master Equation II (Scalar Field Wave Equation)

Varying the Action with respect to the scalar field ($\frac{\delta S}{\delta\psi} = 0$) using Euler-Lagrange equations yields the non-linear wave equation:

$$\Box_j\psi + \nabla_\mu\,(K'(X)\nabla^\mu\psi) = \frac{c^4\psi}{kM_{\text{collective}}}\left(e^{\frac{\pi c T_{\text{local}} d(\hat{\Omega})}{kM_{\text{collective}}}} - 1\right) \tag{10}$$

where $\Box_j \equiv g^{\mu\nu}_j\nabla_\mu\nabla_\nu$ is the d'Alembertian operator evaluated on the $j$-regularized metric.

## III. OBSERVATIONAL BENCHMARKS

### A. Galactic Rotation Curves

In low-acceleration galactic outskirts ($a_N < a_0 \approx 1.2 \times 10^{-10}$ m/s$^2$), spatial gradients in $V(\psi)$ generate an acceleration floor:

$$a_{\text{eff}} = \sqrt{a_N^2 + a_N a_0} \tag{11}$$

This naturally produces flat asymptotic orbital velocity profiles:

$$v_{\text{flat}} = (GMa_0)^{1/4} \tag{12}$$

matching observational galactic rotation curves without requiring cold dark matter particles.

### B. M87* Black Hole Shadow

Calculating null geodesics ($ds^2 = 0$) on the regularized metric $g_{rr} = j$ shifts the photon orbit radius slightly outward. For the supermassive black hole M87*, the framework predicts a 3D shadow angular diameter of:

$$\theta_j \approx 41.3\ \mu\text{as} \tag{13}$$

which lands directly inside the Event Horizon Telescope's observational window of $42 \pm 3$ $\mu$as.

### C. Resolution of the Hubble Tension

In low-density cosmic voids ($z < 0.15$), the mass parameter $M_{\text{collective}}$ drops while void scales $d(\hat{\Omega})$ remain large. Here, $T^{\text{tax}}_{\mu\nu}$ generates a local vacuum energy boost:

$$\Lambda_{\text{eff}}(x^\mu) = \frac{8\pi G}{c^4}V(\psi) \tag{14}$$

This accelerates local spatial expansion to 73.0 km/s/Mpc while seamlessly converging to 67.4 km/s/Mpc at high redshifts ($z > 0.5$).

## IV. HIGH-DENSITY SCREENING MECHANISM

In dense environments such as the Solar System ($M_{\text{collective}} \to$ large):

1. The exponential factor in $V(\psi)$ approaches zero ($e^\epsilon - 1 \approx 0$).
2. The effective mass of field fluctuations $m^2_\psi = \frac{\partial^2 V}{\partial\psi^2}$ becomes extremely heavy.
3. Field gradients decay exponentially ($\nabla_\mu\psi \to 0$), forcing $T^{\text{tax}}_{\mu\nu} \to 0$.

Evaluating the Parameterized Post-Newtonian (PPN) parameter $\gamma$ yields:

$$\gamma \approx 1 - \mathcal{O}(10^{-9}) = 1.000000000 \tag{15}$$

This deviation is three orders of magnitude below the Cassini spacecraft measurement limit ($|\gamma - 1| \leq 2.3 \times 10^{-5}$), proving that the scalar field screen-suppresses itself in local high-density environments.

## V. LIMITATIONS & OPEN QUESTIONS

While the $j$-tax framework successfully reproduces key astrophysical observables, several theoretical aspects remain unresolved and require further investigation:

1. **Quantum Perturbation Stability:** A rigorous check on the field's speed of sound ($c_s^2 \equiv \frac{\partial P}{\partial\rho}$) must be performed to prove the absence of ghost and tachyonic instabilities ($0 \leq c_s^2 \leq c^2$).
2. **CMB Power Spectrum Resolution:** While background expansion tracks $H(z)$, full linear perturbation analysis is required to verify that $K(X)$ reproduces the exact multipole heights ($\ell = 2$ to 2500) of the Planck CMB power spectrum.
3. **Mathematical Topology of $\mathbb{R}[j]$:** The formal differential topological mapping of division-by-zero boundary operators requires rigorous formulation within smooth manifold theory.

## VI. CONCLUSION

We have demonstrated an extended scalar-tensor theory of gravity grounded in $j$-algebra that eliminates horizon infinities, explains flat galactic rotation curves, matches 3D black hole shadow observations, resolves the Hubble Tension, and passes local solar system tests through density-dependent screening.

## DATA & CODE AVAILABILITY

The simulation code supporting the results of this manuscript is freely available alongside this publication record on Zenodo.